

## Employment History Form

### INSTRUCTIONS

*Please answer each question clearly and completely. Do not make any changes to the fields of the form.*

|                |                 |
|----------------|-----------------|
| Last Name      | First Name      |
| <b>MANZATO</b> | <b>AGOSTINO</b> |

*Starting with your present/recent post, list in reverse order every employment you have had.*

*Please use a separate block for each job and remove the block(s) as not required.*

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                 |                                                                                                                                                                |
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| <b>WORK EXPERIENCE 1:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                 |                                                                                                                                                                |
| From (DD/MM/YYYY)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | To (DD/MM/YYYY) | Job title                                                                                                                                                      |
| <b>01/07/2000</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>PRESENT</b>  | <b>RESEARCHER AT OSMER – ARPA FVG<br/>(OSSERVATORIO METEOROLOGICO REGIONALE<br/>DELL'AGENZIA PER LA PROTEZIONE DELL'AMBIENTE<br/>DEL FRIULI VENEZIA GIULIA</b> |
| Name of employer<br><b>ARPA FVG</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 | Location (City and Country)<br><b>PALMANOVA, Italy</b>                                                                                                         |
| Current Annual Salary (approx.)<br><b>36000</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                 | Number of direct reports                                                                                                                                       |
| UN Grade/Level (if applicable)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                 |                                                                                                                                                                |
| <p>Main responsibilities:</p> <p>I'm responsible for the scientific research made at the regional met office, which mainly regard studies on convection. I have studied the local climatologies of rain (from raingauges), lightning (from the european network of Vaisala sensors) and hail (from hailpad network), including their trends. I have studied case studies of strong severe events, including supercells, bow echoes, strong hailstorms. I have developed forecasting methods based on artificial intelligence (e.g. neural networks) for forecasting lightning, rain and hail, both from instability indices derived from the analysis of the atmospheric profile than from statistical downscaling of ECMWF IFS model forecast. I have made also theoretical studies regarding forecast verification issues. Recently I collaborated with dr Jun-Ichi Yano (MeteoFrance) to study cloud physics, as the relation between atmospheric precipitable water and rain at surface.</p> <p>In 2017 I was tutor of a student during his thesis work on hailpads analysis in the frame of the bachelor degree in Environmental Sciences at the University of Udine – Italy. In 2018 and 2019 I was tutor of two students of the Udine University (Department of Mathematics, Computer Science and Physics) during their thesis on thunderstorm forecasting with convolutional neural networks (CNN) and analogue methods (AtmoSwing).</p> |                 |                                                                                                                                                                |

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Main achievements:

My main achievements are:

- 1) forecasting programs operationally used by forecasters working at OSMER. These programs are based on Neural Networks, Convolutional Neural Networks (developed in Pytorch) and genetic algorithms. All my programs are written in Python and or R.
- 2) participating at international conferences to show the results of our studies and to be updated on latest findings regarding severe convection. Since 2009 I was always invited to participate to all the Convection Working Group Workshops supported by EUMETSAT. In 2017 I was asked to become member of the scientific committee of the Second European Hail Workshop (Univ. Bern, CH).
- 3) Publishing the main results of our research, made also in collaboration with important researches from all around the globe, in international peer review journals. Example of the most recent (2016–2022) publications are the following:

Manzato, A., Cicogna, A., Centore, M., Battistutta, P., & Trevisan, M. (2022). Hailstone characteristics in NE Italy from 29 years of hailpad data, *Journal of Applied Meteorology and Climatology* <https://doi.org/10.1175/JAMC-D-21-0251.1>

Manzato, A., Serafin, S., Miglietta, M. M., Kirshbaum, D., & Schulz, W. (2022). A pan-Alpine climatology of lightning and convective initiation, *Monthly Weather Review* <https://doi.org/10.1175/MWR-D-21-0149.1>

Yano, J., & Manzato, A. (2022). Does More Moisture in the Atmosphere Lead to More Intense Rains?, *Journal of the Atmospheric Sciences*, 79(3), 663–681. <https://doi.org/10.1175/JAS-D-21-0117.1>

Pucillo, A., Miglietta, MM, Lombardo, K, Manzato, A. (2020) Application of a simple analytical model to severe winds produced by a bow echo like storm in northeast Italy. *Meteorol Appl.* <https://doi.org/10.1002/met.1868>

Jl Yano, MHP Ambaum, HF Dacre, A Manzato (2020) [A dynamical-system description of precipitation over the tropics and the midlatitudes](https://doi.org/10.1080/16000870.2020.1847939), *Tellus A: Dynamic Meteorology and Oceanography* 72 (1), 1–17 <https://doi.org/10.1080/16000870.2020.1847939>

Manzato, A, Riva, V, Tiesi, A, Marcello Miglietta, M. (2020) Observational analysis and simulations of a severe hailstorm in northeastern Italy. *QJR Meteorol Soc.* 2020; 146: 3587–3611. <https://doi.org/10.1002/qj.3886>

A. Manzato, A. Pucillo, A. Cicogna, (2019) Improving ECMWF-based 6-hours maximum rain using instability indices and neural networks, *Atmospheric Research*, 217, 184–197.

<https://doi.org/10.1016/j.atmosres.2018.10.020>

Martius, O., Hering, A., Kunz, M., Manzato, A., Mohr, S., Nisi, L., & Trefalt, S. (2018). Challenges and Recent Advances in Hail Research, *Bulletin of the American Meteorological Society*, 99(3), ES51–ES54. <https://doi.org/10.1175/BAMS-D-17-0207.1>

Ukkonen, P., Manzato, A., & Mäkelä, A. (2017).

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Evaluation of Thunderstorm Predictors for Finland Using Reanalyses and Neural Networks, *Journal of Applied Meteorology and Climatology*, 56(8), 2335–2352.

<https://doi.org/10.1175/JAMC-D-16-0361.1>

Manzato, A. and Jolliffe, I. (2017)

Behaviour of verification measures for deterministic binary forecasts with respect to random changes and thresholding. *Q.J.R. Meteorol. Soc.*, 143: 1903–1915.

<https://doi.org/10.1002/qj.3050>

Miglietta, M.M., Manzato, A. and Rotunno, R. (2016)

Characteristics and predictability of a supercell during HyMeX SOP1. *Q.J.R. Meteorol. Soc.*, 142: 2839–2853.

<https://doi.org/10.1016/j.atmosres.2018.10.020>

Davolio, S., Volonté, A., Manzato, A., Pucillo, A., Cicogna, A. and Ferrario, M.E. (2016),

Mechanisms producing different precipitation patterns over north-eastern Italy: insights from HyMeX-SOP1 and previous events. *Q.J.R. Meteorol. Soc.*, 142: 188–205. <https://doi.org/10.1002/qj.2731>

Mecikalski, J. R., Rosenfeld, D., and Manzato, A. (2016)

Evaluation of geostationary satellite observations and the development of a 1–2 h prediction model for future storm intensity, *J. Geophys. Res. Atmos.*, 121, 6374–6392,

doi:[10.1002/2016JD024768](https://doi.org/10.1002/2016JD024768).

A. Manzato, A. Cicogna, A. Pucillo, (2016)

6-hour maximum rain in Friuli Venezia Giulia: Climatology and ECMWF-based forecasts, *Atmospheric Research*, 169, 465–484,

<https://doi.org/10.1016/j.atmosres.2015.07.013>.

### WORK EXPERIENCE 2:

| From (DD/MM/YYYY)                               | To (DD/MM/YYYY) | Job title                                                                                                     |
|-------------------------------------------------|-----------------|---------------------------------------------------------------------------------------------------------------|
| 20/11/1996                                      | 30/06/2000      | <b>Researcher at the Thunderstorm and Hail Prevention Research Center of Centro Servizi Agrometeorologici</b> |
| Name of employer<br><b>CSA</b>                  |                 | Location (City and Country)<br><b>Cervignano del Friuli, Italy</b>                                            |
| Current Annual Salary (approx.)<br><b>18000</b> |                 | Number of direct reports                                                                                      |
| UN Grade/Level (if applicable)                  |                 |                                                                                                               |

#### Main responsibilities:

I was responsible for the data format conversion and visualization of all meteorological dataset (stations, soundings, lightning, hailpads, satellite, radar,...) using the NCAR Zebra suite.

I worked under the supervision of prof. Griffith Morgan on the hail project, including the management of the FVG hailpad network and data analysis.

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### Main achievements:

I developed a program (Sound\_Analysis.PY) to evaluate the atmospheric instability, computing more than 50 indices. That program was the base for the following studies on thunderstorms, that later culminated in the publications of my first papers, like for example:

A. Manzato, G. Morgan, (2003)

Evaluating the sounding instability with the Lifted Parcel Theory, *Atmospheric Research*, 67–68, 455–473,  
[https://doi.org/10.1016/S0169-8095\(03\)00059-0](https://doi.org/10.1016/S0169-8095(03)00059-0)

A. Manzato, (2003)

A climatology of instability indices derived from Friuli Venezia Giulia soundings, using three different methods, *Atmospheric Research*, 67–68, 417–454,  
[https://doi.org/10.1016/S0169-8095\(03\)00058-9](https://doi.org/10.1016/S0169-8095(03)00058-9)

Manzato, A. (2005).

The Use of Sounding-Derived Indices for a Neural Network Short-Term Thunderstorm Forecast, *Weather and Forecasting*, 20(6), 896–917.  
<https://doi.org/10.1175/WAF898.1>

Manzato, A. (2005).

An Odds Ratio Parameterization for ROC Diagram and Skill Score Indices, *Weather and Forecasting*, 20(6), 918–930.  
<https://doi.org/10.1175/WAF899.1>

During the initial period prof. Morgan sent me to visit important Institutions (including NCAR, NSSL, SPC) to supply at my lack of PhD. After that, a part from participating to the most important conferences (European Conference on Severe Storms, International Conference on Alpine Meteorology, European Hail workshop,...), in my career I followed made a lot of post-doc courses/summer schools, including also:

Summer colloquium on the physics of the weather and climate “The effect of the topography on the atmospheric circulation” ICTP Trieste June 1998

Summer Colloquium on Ice formation in the Atmosphere (featuring also prof. W. Al Cooper) NCAR Boulder June 1999

Grand Combin Summer School on Physics and Predictability of Rainfall and Floods (featuring also prof. K. Emanuel) ISAC Saint Oyen July 2002

Summer School on Mountain Meteorology "Orographic Effects on Precipitation" (featuring also dr. R. Rotunno) Univ. Trento Trento July 2004

CISM Advanced School on Atmospheric Convection: Research and Operational Forecasting Aspects (featuring also prof. P. Markowsky) CISM Udine July 2005

Short Course on Artificial Intelligence Applications to Environmental Science (featuring also prof. C. Marzban) AMS Atlanta January 2006

EUMETSAT Training Workshop on Interpretation of Satellite Images/Products of Convective Storms (featuring also prof. D. Rosenfeld) ICTP Trieste September 2007

Third Mediterranean School on Mesoscale Meteorology: "Fundamentals and applications of probabilistic forecasts and Ensemble Prediction Systems" (featuring also dr. T. Hamill) ISAC Alghero May 2008

Conference on Teleconnections in the Atmosphere and Oceans ICTP Trieste November 2008

Fourth ISAC Summer School: “Severe Convection Weather: Theory and Applications” (featuring also prof. E. Fedorovich) ISAC Castro Marina September 2012

The Fluid Dynamics of Climate (featuring also prof. M. Ghil) CISM Udine August 2013

Sixth ISAC Summer School: “Advances in severe weather analysis: models and observations” (featuring also prof. H. Bluestein) ISAC Castro Marina June 2016

Numerical Modeling, Predictability and Data Assimilation in Weather, Ocean and Climate ISAC Bologna

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October 2017

That is why I believe that my formation is at least equivalent to that of a PhD.

| WORK EXPERIENCE 3:              |                 |                             |
|---------------------------------|-----------------|-----------------------------|
| From (DD/MM/YYYY)               | To (DD/MM/YYYY) | Job title                   |
|                                 |                 |                             |
| Name of employer                |                 | Location (City and Country) |
| Current Annual Salary (approx.) |                 | Number of direct reports    |
| UN Grade/Level (if applicable)  |                 |                             |
| Main responsibilities:          |                 |                             |
| Main achievements:              |                 |                             |